EXHAUST SHUTTER VALVE [SKYACTIV-D 1.5]

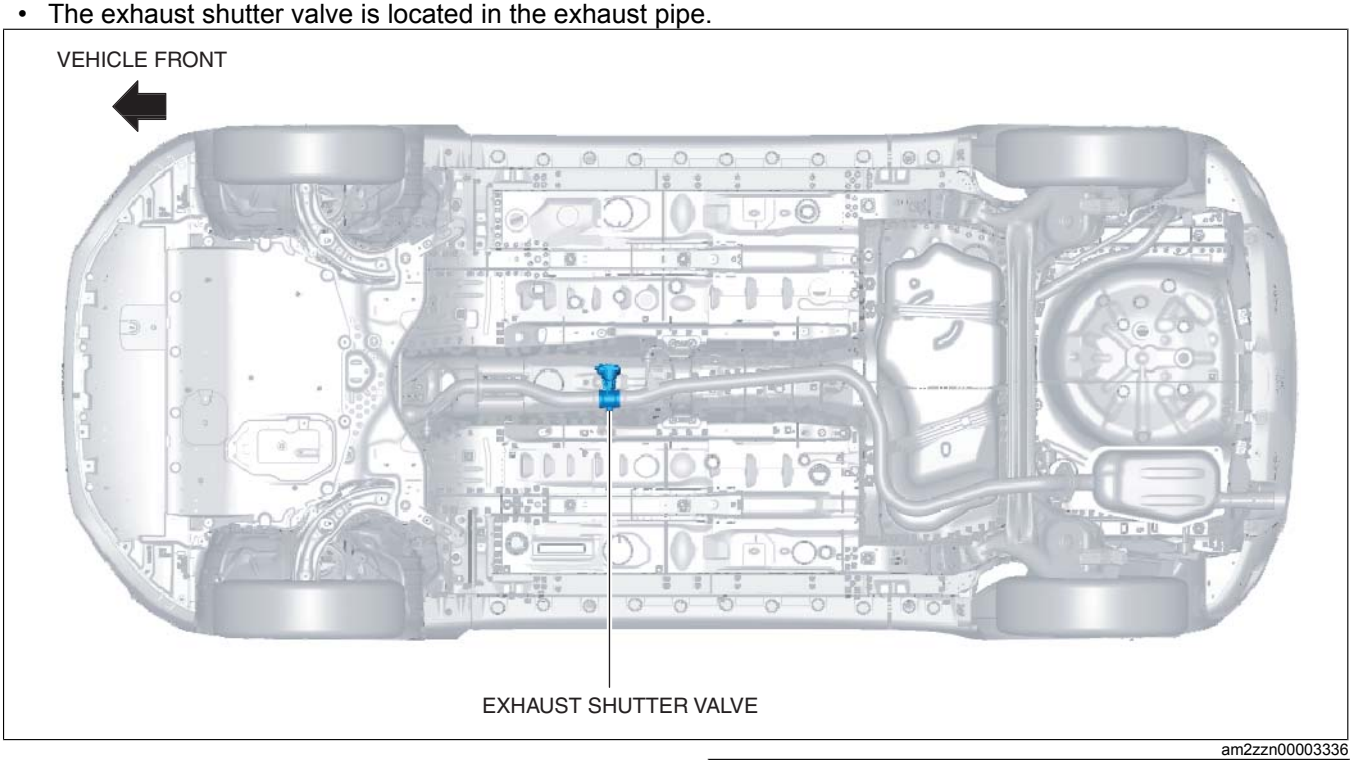
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Purpose, Function

- The exhaust shutter valve adjusts the exhaust gas volume recirculated to the EGR cooler side to adjust the exhaust gas volume released to the atmosphere.

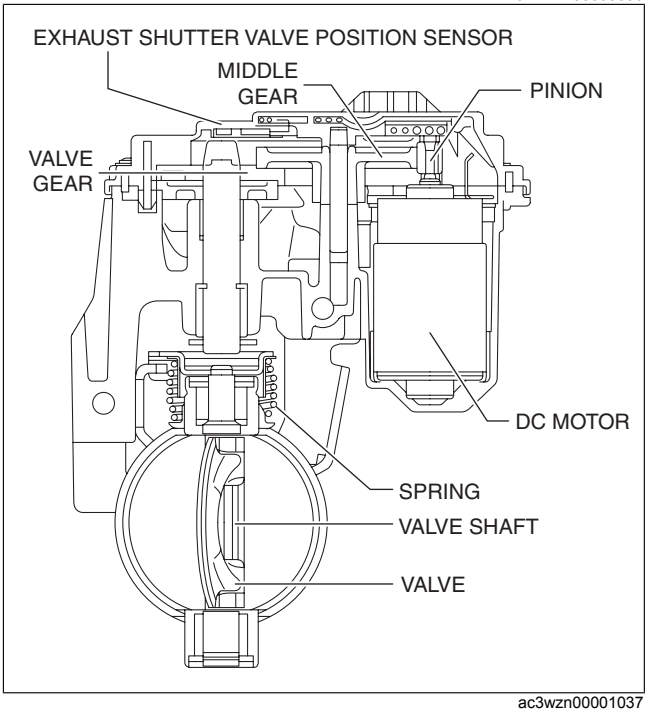
Construction

- The exhaust shutter valve is located in the exhaust pipe.



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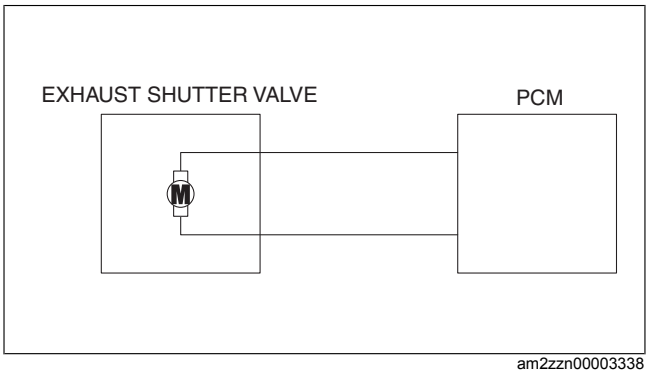
- The exhaust shutter valve consists of the exhaust shutter valve position sensor, valve gear, mid gear, pinion, DC motor, spring, valve shaft, and valve.
- The exhaust shutter valve has an integrated exhaust shutter valve position sensor. For details on the exhaust shutter valve position sensor, refer to the CONTROL SYSTEM. (See EXHAUST SHUTTER VALVE POSITION SENSOR [SKYACTIV-D 1.5].)



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Operation

- The exhaust shutter valve is opened to the valve opening angle according to the energization current flowing to the DC motor from the PCM.
- The exhaust shutter valve performs coordinated control with the intake shutter valve, LP-EGR control valve, and HP-EGR control valve. For details on the exhaust shutter valve control, refer to the CONTROL SYSTEM. (See EXHAUST SHUTTER VALVE CONTROL [SKYACTIV-D 1.5].)
- In the event that the exhaust shutter valve has a malfunction and the DC motor control is disabled, the shutter valve is opened to the fully open position by the return spring. Due to this, the air amount necessary to keep the vehicle moving is assured.



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Fail-safe

DTC No.	Fail-safe function
P0475:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
P0476:00	
P049C:00	
P048D:00	• Limits the engine torque. • Inhibits the EGR control.
P048E:00	
P049B:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits engine-stop by operating the i-stop function.